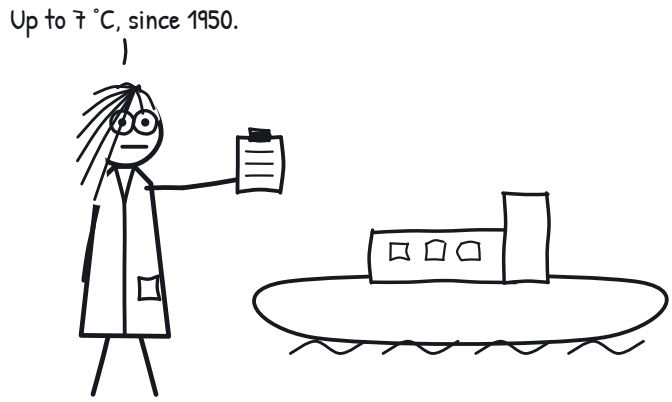
Ice, Plankton and Carbon: Five Januaries on the Western Antarctic Peninsula

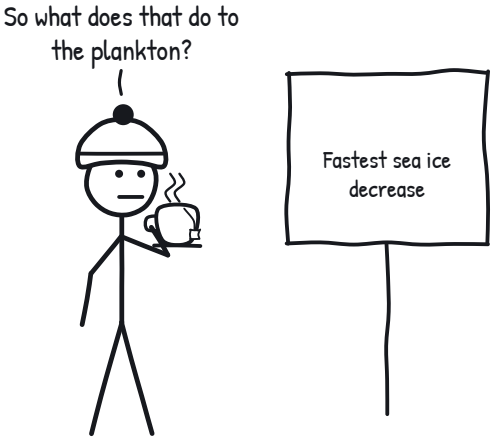
From "Decline in plankton diversity and carbon flux with reduced sea ice extent along the Western Antarctic Peninsula" by Yajuan Lin, Carly Moreno, Adrian Marchetti, Hugh Ducklow, Oscar Schofield, Erwan Delage, Michael Meredith, Zuchuan Li, Damien Eveillard, Samuel Chaffron, Nicolas Cassar (2021)

The fastest-changing coast

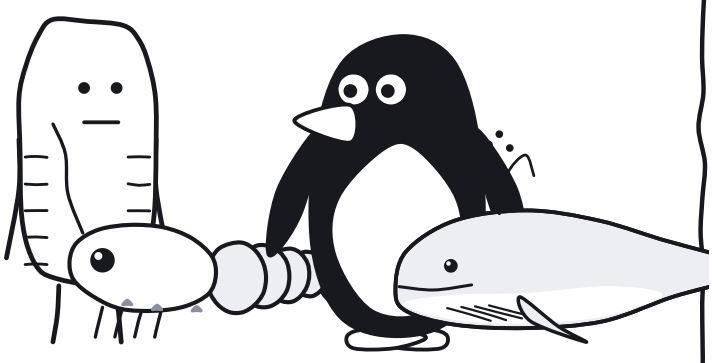
The Western Antarctic Peninsula system has exhibited some of the most significant changes in the Southern Ocean, with rising air temperature up to 7 °C since 1950.



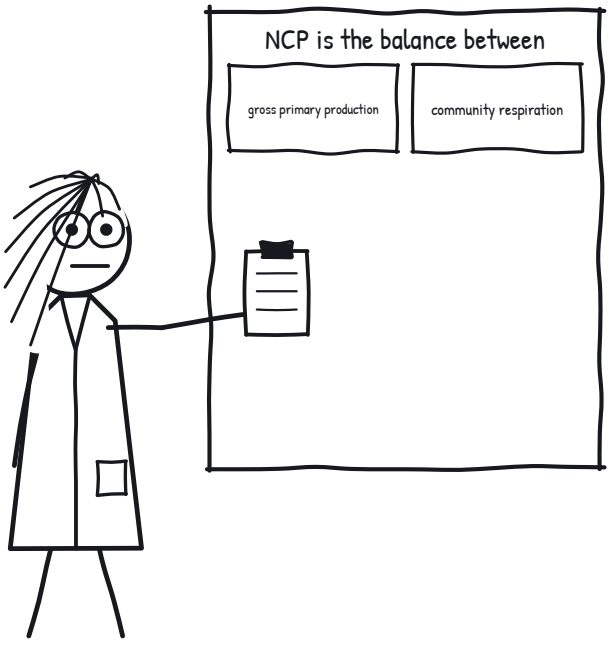
The Western Antarctic Peninsula has seen the fastest sea ice decrease in Antarctica.



Eukaryotic plankton at the Western Antarctic Peninsula, including phytoplankton and microzooplankton, support higher trophic levels ranging from krill to penguins and whales.



The net community production (NCP) represents the balance between gross primary production and community respiration.



When the organic carbon pool in the mixed layer is in a steady state, NCP reflects the organic carbon available for export out of the surface mixed layer.

So NCP tells us how much carbon is available to leave the surface?



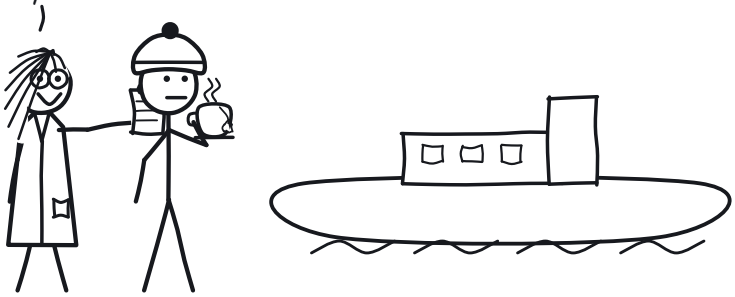
Given the fundamental importance of eukaryotic plankton at the WAP, it is imperative to understand and predict the changes in plankton community structure, biodiversity, and carbon flux in this rapidly changing environment.



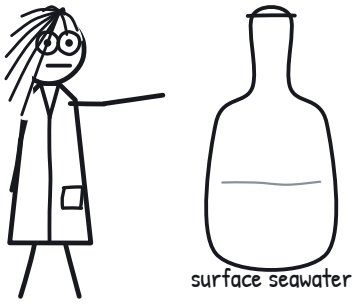
Five Januaries at sea

Each year in January, a research vessel conducted intensive oceanographic and biological surveys across the shelf-transects and a north-south gradient at the West Antarctic Peninsula.

Five years: 2012 to 2016.

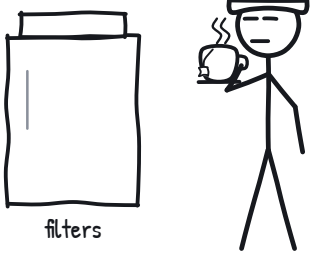


Surface seawater from the ship's underway flow-through system was gently vacuum-filtered onto a Supor filter.



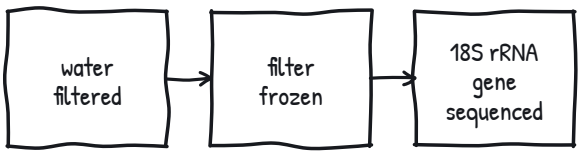
The filters were immediately stored at −80 °C until further analysis.

My tea is nowhere near that cold.

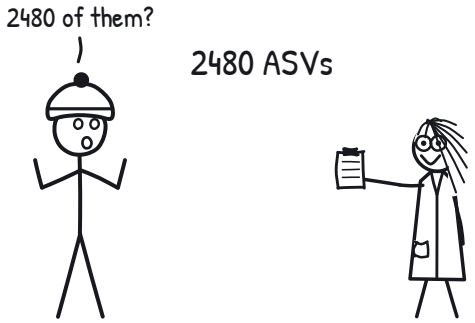


Based on a five-year WAP DNA sampling, the authors explored the plankton community structure and diversity via sequencing of the 18S rRNA gene marker.

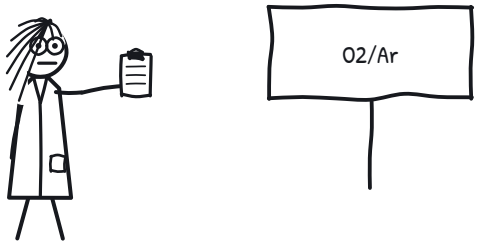




At the finest taxonomic resolution, 2480 amplicon sequence variants (ASVs) were identified from the five-year amplicon dataset.



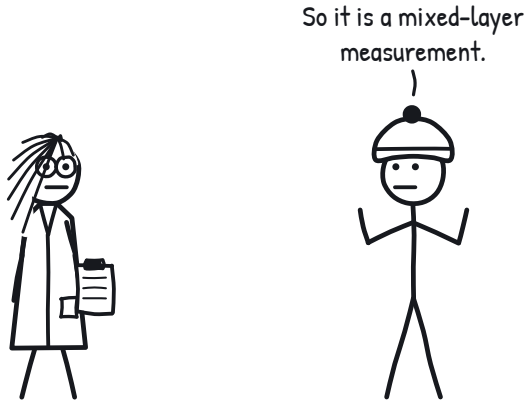
Seawater O2/Ar ratios were measured underway from the ship's flow-through system, using an equilibrator inlet mass spectrometer (EIMS).



Using Ar, an inert gas with similar solubility properties as O2, the authors decomposed total O2 into physical and biological components.

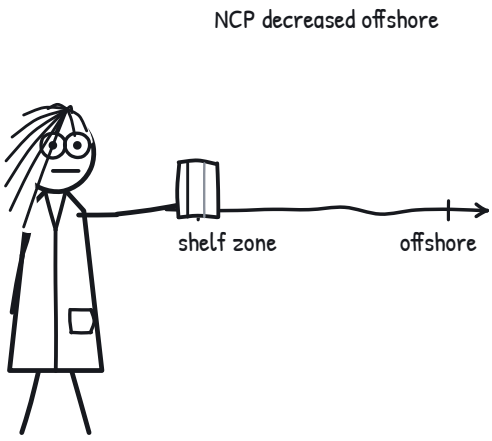


The O2/Ar-NCP measurements in this study only reflect the mixed layer carbon fluxes, and the authors do not assess the sequestration timescales.

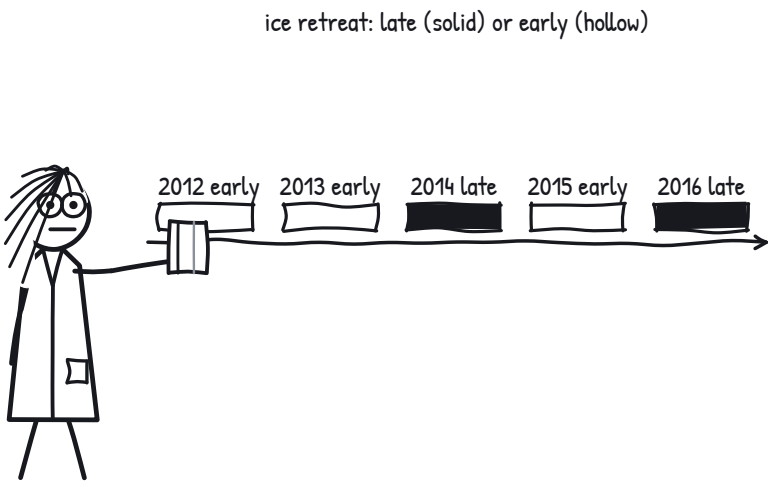


Who turns up in which year

During the summer, NCP was highest in the shelf zone and decreased offshore.



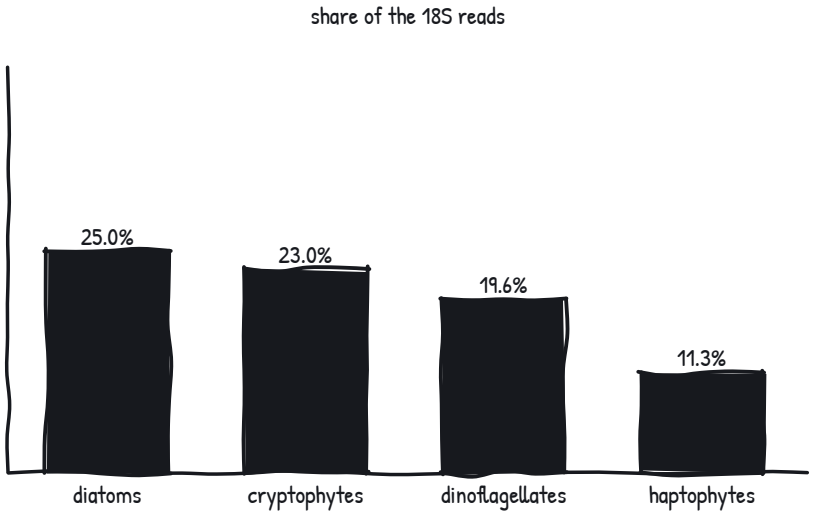
The two years that feature late sea-ice retreat (2014 and 2016) were associated with abnormally high summer NCP (t-test, $p < 0.0001$).



The five years studied (2012 to 2016) were a period of moderately positive sea-ice area anomalies following a more prolonged period of anomalously low sea-ice area.

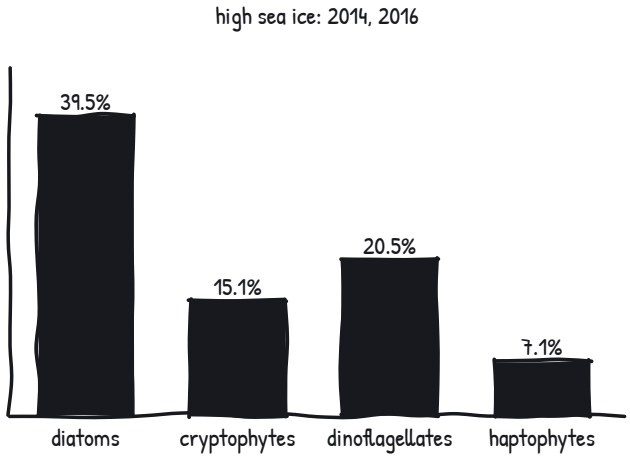


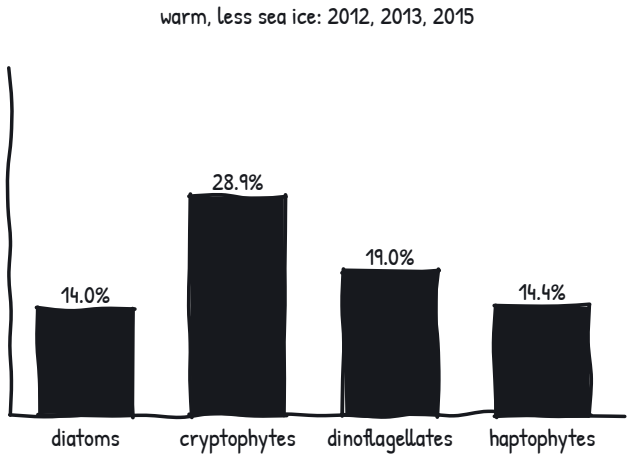
Four eukaryotic plankton groups dominated the WAP surface water: diatoms (25.0%), cryptophytes (23.0%), dinoflagellates (19.6%) and haptophytes (11.3%).



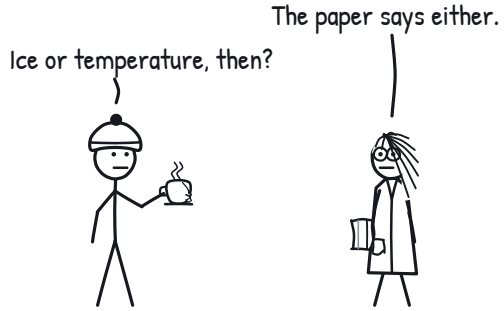
In the high-sea-ice years (2014 and 2016) diatoms were on average 39.5 ± 3.0% of the community; in the warm years with less sea ice (2012, 2013 and 2015), 14.0 ± 1.7%.

Cryptophytes went the other way: 15.1 ± 6.1% against 28.9 ± 8.3%.

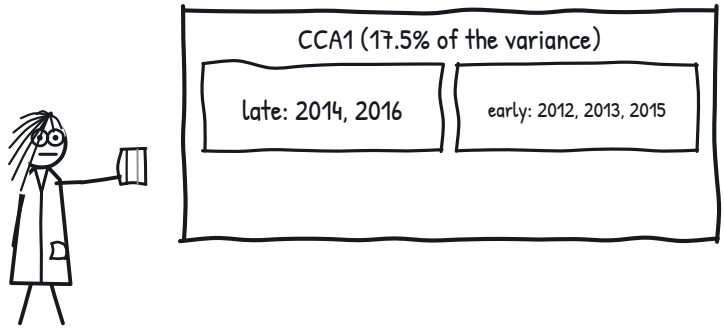




Canonical correspondence analysis (CCA) illustrates that either ice conditions or SST is the dominant driver on community structure at the ASV level.



The first axis CCA1 (17.5% of the variance) separates samples from late (2014 and 2016) and early (2012, 2013, and 2015) ice-retreat years.



The most substantial abiotic factor associated with CCA1 is SST (negatively correlated).

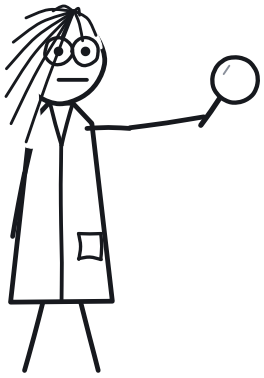


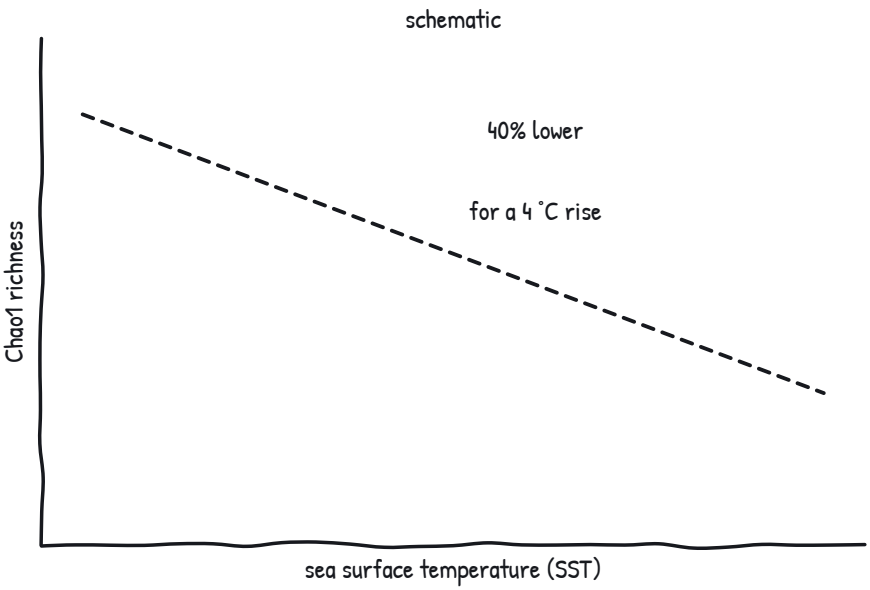
Overall, sea-ice conditions and associated environmental parameters, such as low SST and low salinity, are the primary drivers of community differentiation at the ASV level.



Warmer water, less diversity

Chao1, a measure for species richness, demonstrated an evident decline towards higher SST, with a 40% decrease in the index for a 4 ˚C rise in SST.





Pielou's evenness and Shannon, which consider both species richness and evenness, also decreased significantly with increasing SST.

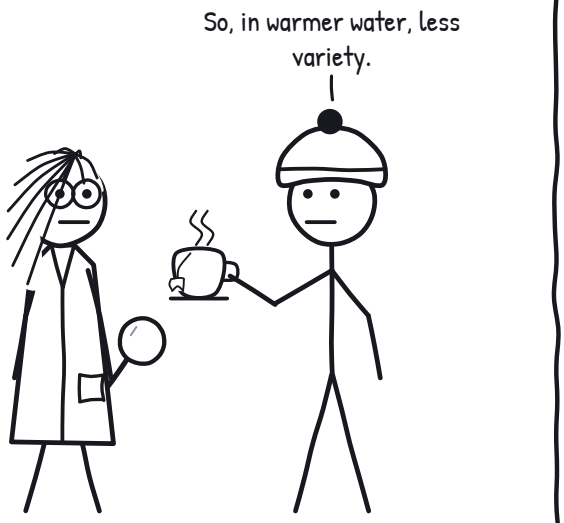




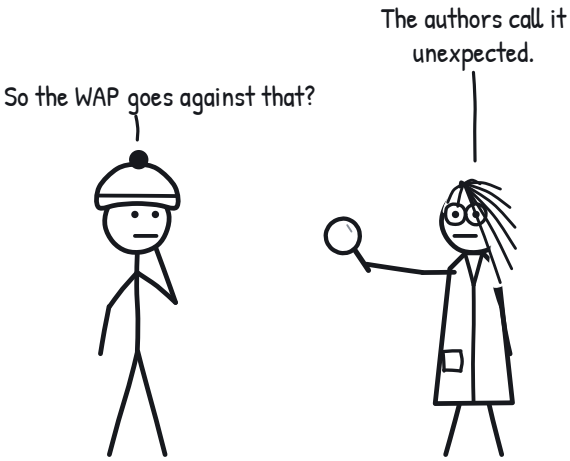
All three diversity indices show significant negative correlations with SST (two-sided t-test, $p < 0.01$).



Communities in warmer WAP waters show lower richness and lower evenness, i.e., a few taxa dominate.



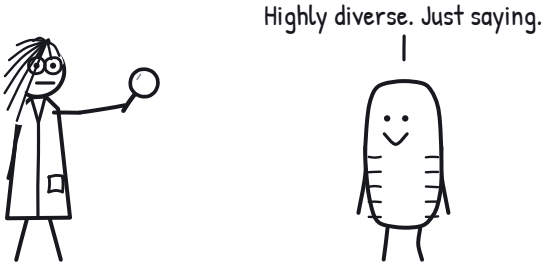
In a recent global analysis from Tara Oceans, temperature was also the major explanatory factor for eukaryotic plankton biodiversity, but with the opposite trend: lower diversity towards higher latitude or lower temperature.



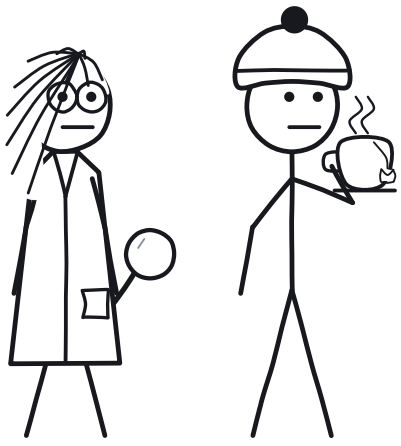
The Tara Oceans dataset featured only limited sampling in the Southern Ocean (three data points).



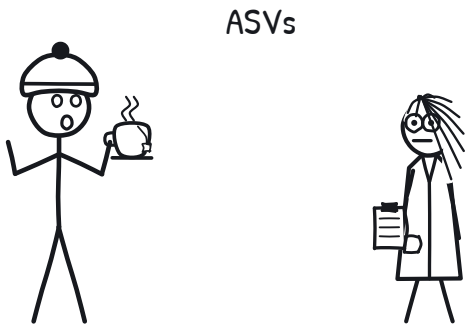
One explanation: the ice-associated plankton communities consist mainly of diatoms, which are highly diverse and can thrive at lower temperatures compared to other phytoplankton.



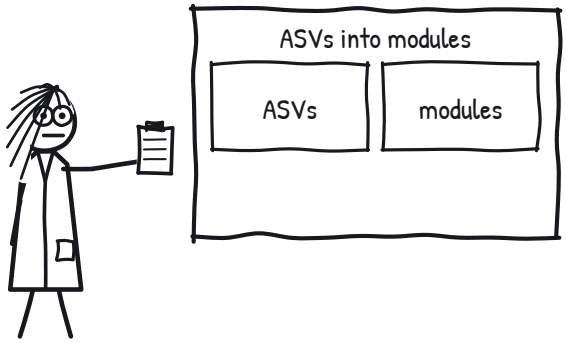
The authors write: "our results suggest that global warming may decrease plankton diversity in coastal Antarctica."



Due to the high dimensionality of the ASV dataset, it is challenging to model NCP based on community structure.



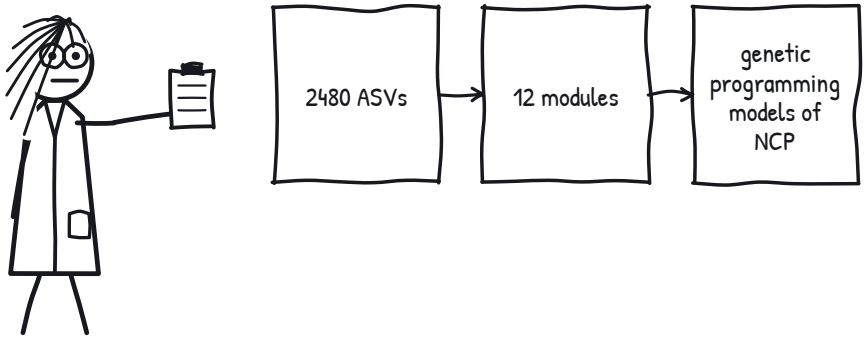
The authors applied a weighted gene correlation network analysis (WGCNA) approach to delineate clusters of 18S ASVs into subnetworks or modules.



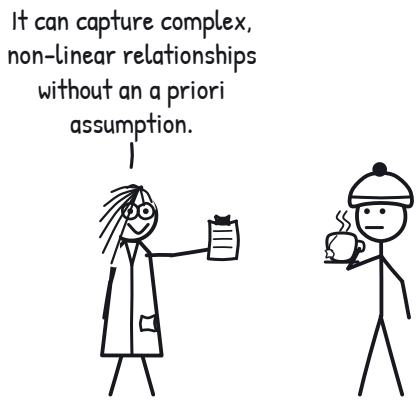
Each module represents an assemblage of predicted highly interconnected plankton community members, potentially indicating a group of organisms with strong ecological overlap and/or interactions.



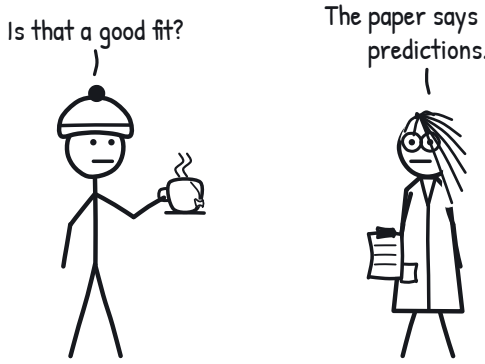
In total, the authors identified 12 modules from the five-year global community structure.



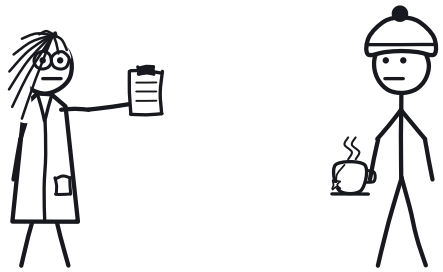
Genetic Programming (GP) is a machine learning approach based on evolution computation.



The top four GP solutions provide good predictions on NCP with R2 ranging from 0.70 to 0.80.



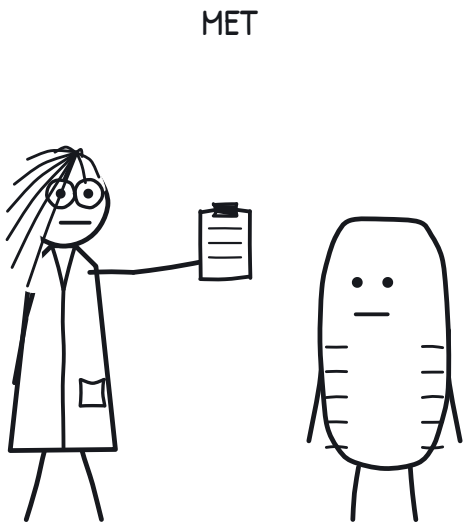
The top two physical factors ranked by selection frequency are SST and mixed layer depth (MLD), followed by surface photosynthetically active radiation (PAR).



This suggests that temperature and light are likely the primary physiological limiting factors on NCP in the WAP system.



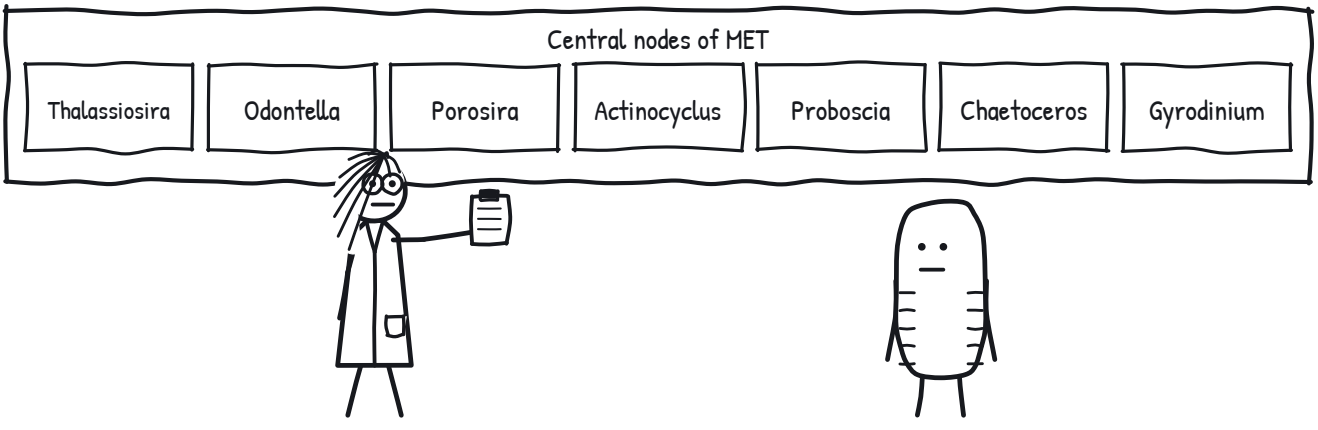
Among the community-assemblage factors, module turquoise (MET) is the most important predictor for NCP.



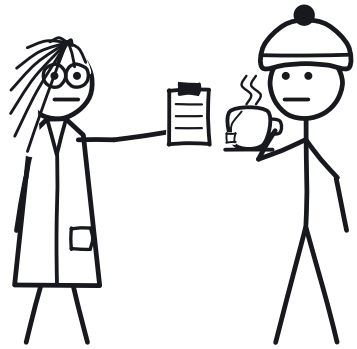
MET is also the largest module identified, which consists of 126 ASVs, mainly representing diverse groups of diatoms and dinoflagellates.



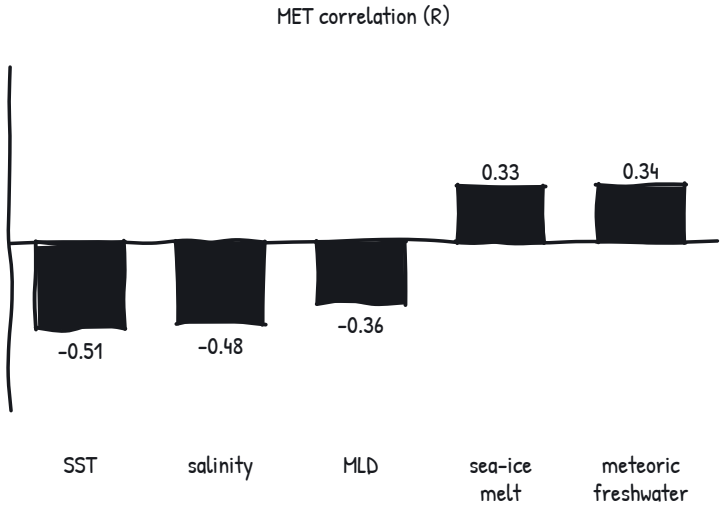
The central nodes in the MET network, the top-10 most connected ASVs, include the diatom genera and the dinoflagellate Gyrodinium.



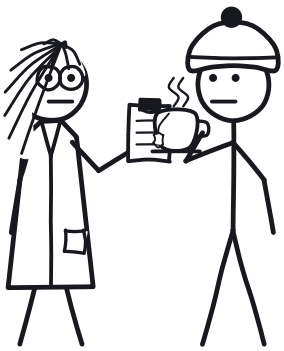
MET appears in all top-four performed GP solutions, and it is the sole biological factor in solution 1 explaining a majority of the NCP variability (R2 = 0.70).



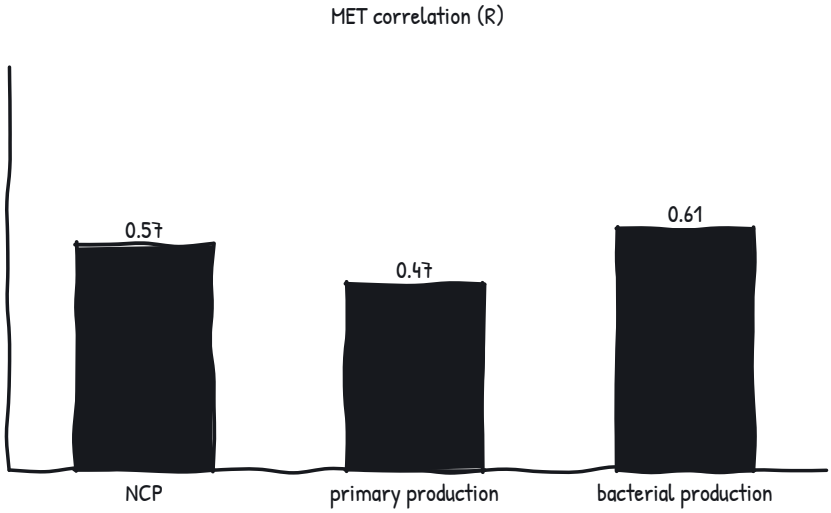
MET is significantly correlated with low SST, low salinity, shallow MLD, elevated sea-ice melt and meteoric freshwater.



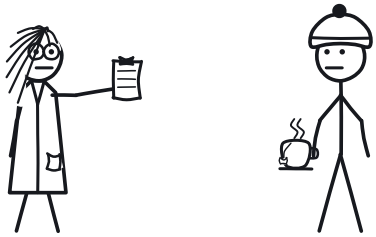
The overall spatial distributions of MET are consistent with the January averaged sea-ice distribution estimated by satellite.



NCP, ship-based primary production and bacterial production are all positively correlated with MET. This indicates that MET-dominated regions have high biological activities.



Two other community-assemblages, MER (module red, 27 ASVs) and MEG (module green, 29 ASVs), also contribute to NCP models but to a lesser extent.



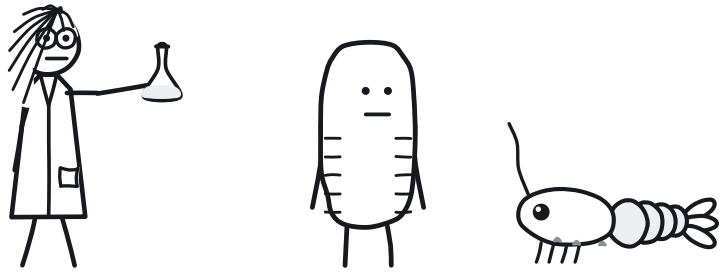
MER is dominated by the cryptophyte Geminigera, which appears in warmer waters. MEG, heterotrophic protists dominated by Picomonas and Telonema, contributes to NCP negatively in two GP solutions.



MER: warmer water

MEG: heterotrophic protists

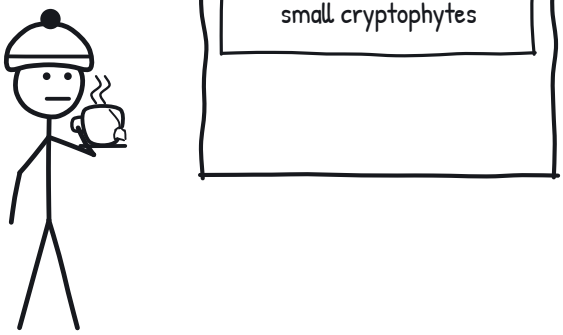
The authors hypothesize three broad patterns. (i) Large centric diatoms associated with ice-melt form intensive blooms and fuel a short food chain from krill to other top predators.



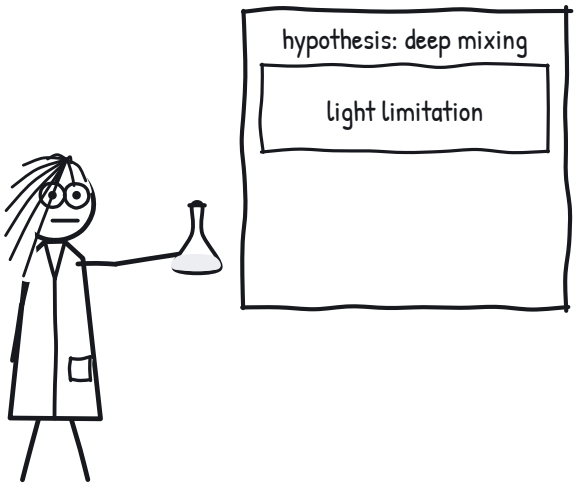
On this hypothesis, high productivity combined with small losses through trophic transfer results in high export production.



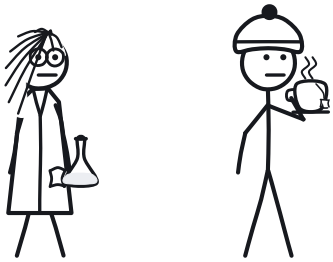
(ii) In warmer water, small cryptophytes dominate. Compared to large diatoms, their growth could be more efficiently checked by small microzooplankton grazers, thus resulting in lower biomass for export.



(iii) With deep mixing, primary production in the water column is low due to light limitation. The last pattern displays the lowest NCP.



With climate change, the WAP region is projected to have a significant loss in summer sea ice, a rise in sea surface temperatures, and deeper mixing associated with more open water and stronger winds.

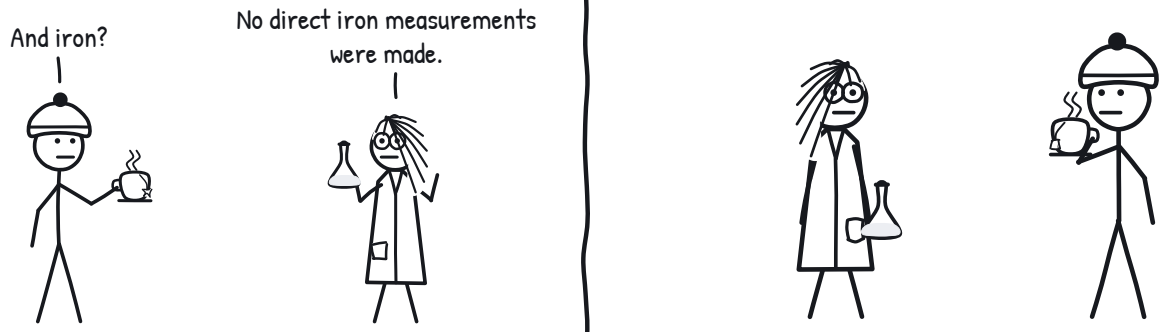


Consequently, the latter two scenarios may become more prevalent in the upcoming years to decades.



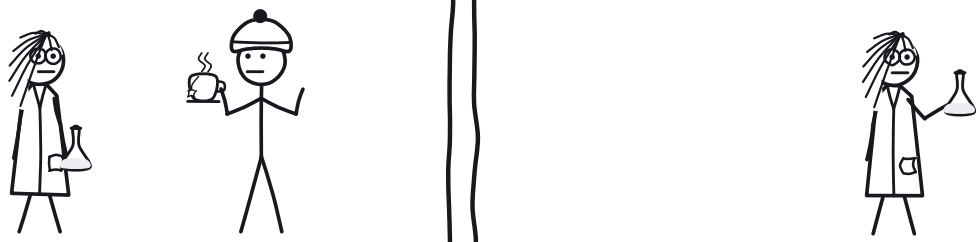
Because iron concentrations were not included in this analysis, the authors cannot discern whether sea ice and/or glacial ice melt impact productivity through altering light and/or iron levels.

The study represents the longest record of eukaryotic DNA-based community structure and NCP in coastal Antarctica, but its observations are limited to seasonal snapshots of the (summer) WAP system.

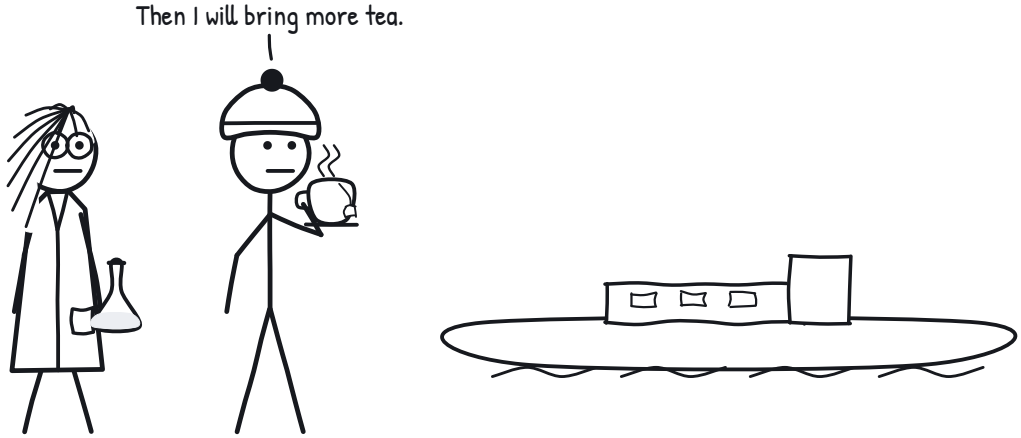


Correlation-based analyses and statistical models will need to be further validated with field incubations and lab experiments.

To the extent that interannual variability in the influence of sea ice extent serves as a proxy for longer-term consequences of climate change, the WAP and other coastal Antarctica regions could be destined for reduced biodiversity and biological carbon drawdown.



However, a longer time series will be needed to confirm the pattern.



Adapted from "Decline in plankton diversity and carbon flux with reduced sea ice extent along the Western Antarctic Peninsula" by Yajuan Lin, Carly Moreno, Adrian Marchetti, Hugh Ducklow, Oscar Schofield, Erwan Delage, Michael Meredith, Zuchuan Li, Damien Eveillard, Samuel Chaffron, Nicolas Cassar (2021), doi 10.1038/s41467-021-25235-w. Licence: CC BY 4.0. Condensed into a comic with invented characters and scenes using the science-comic-pipeline; the paper's authors did not write or endorse this document.

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